

COTHROM Press Release

The Problem Solving Association CLG.

Could the rules for redrawing Dáil constituencies actually deliver on what they promise?

New Irish research applies statistical physics to Ireland's electoral map, making the trade-offs behind constituency boundaries visible for the first time

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Redrawing a constituency map is like completing a jigsaw puzzle with no picture on the box. The pieces are electoral divisions (EDs), there is no single correct way to arrange them, and every arrangement trades one goal off against another: equal representation, respect for county lines, sensible shapes, and continuity with the map voters already know.

A new piece of research from The Problem Solving Association CLG. (TPSA), conducted by a team of mathematicians and physicists stemming as current and past students and researchers from Trinity College Dublin, attempts to make those trade-offs visible instead of leaving them buried inside one proposed map. Following recent coverage of the work in The Irish Times, the team is now setting out the research and its implications directly for the public.

The project, released as a research paper under the name COTHROM (Constituency Optimisation Through Hamiltonian Representation Of Mandates), borrows ideas from statistical physics. Instead of judging one map at a time, the model scores millions of possible constituency arrangements at once, treating badly-balanced maps as having “high energy” and well-balanced maps with “low energy” - the same framework physicists use to describe a system settling into its most stable state. Two computational search methods working in unison, Markov chain Monte Carlo and simulated annealing, work through an enormous number of configurations in minutes, sometimes intentionally considering poor maps in its early stages to avoid settling for a mediocre result too early.

Tested as a proof of concept on Co. Cork constituencies, COTHROM produced maps with population variances under 0.5% - a result that held across a range of different priority weightings, not just for one favourable setting - compared to 2.2 - 5.6% under the redraw of the same constituencies by An Coimisiún Toghcháin (The Electoral Commission) in their 2023 Constituency Review. The team is upfront that this is early-stage work; it doesn't yet account for each and every consideration made by An Coimisiún when redrawing constituencies, such as regional geographies or local community ties, although they discuss paths to building these in.

“We have seen internationally what distrust in the electoral process can do in terms of societal polarisation. COTHROM, meaning “fair” in Irish, is an algorithmic tool built for Irish electoral redistricting, structured using the rigour of advanced statistical physics methods. This work is a step towards making Irish redistricting completely transparent and as representative as is mathematically possible, earning the trust which is vital to a functioning democracy.”

Michael A.J. Mitchell, CEO of The Problem Solving Association CLG

“I started working with some of these computational methods during my bachelor's degree but I never thought I'd be applying them in such a real-world use case. I've been able to apply skills from my PhD research, ranging from technical programming to project management - even if the context of constituency redistricting isn't too similar to theoretical & computational particle physics! I enjoy the work and discussions with my fellow researchers through the Association, and am excited to scale up our framework for use by legislators, statutory bodies, and the general public alike.”

Ruaidhrí Campion, COTHROM Technical Lead



The Constitution of Ireland requires one Teachta Dála for every 20,000 to 30,000 people on average, and legislation directs An Coimisiún Toghcháin to balance population equality, county boundaries, geographical features, and continuity with existing maps across constituencies with three, four, or five seats. With the population still growing and the next census set for 9 May 2027, the team argues that the following questions are only going to get more relevant as time goes on:

- Is there room to consider if six-seat constituencies could be implemented?
- Does protecting county boundaries always mean accepting more population inequality?
- Are there valid alternative maps that do at least as well as an official proposal on every measure that matters?
- How would the Electoral Commission enforce their qualitative understanding of what makes a “good” constituency and their relative importance?
- How can we ensure that the electorate is in an informed position regarding the nature of how their constituencies change?

The researchers make clear that COTHROM is built to support judgement, not replace it. What it does is generate credible alternatives quickly, make the assumptions behind them reproducible, and show plainly what is being weighed against what - so that whoever draws the next set of boundaries can see the whole shape of the trade-off, not just one version of it.

About The Problem Solving Association. The Problem Solving Association CLG. (TPSA) is an Irish non-profit applying mathematical and technological tools to public-service problems, while staying just as focused on whether those tools should be used and whom they affect. Its work spans electoral systems, public infrastructure, and educational access, and is built through open-source development and transparent methodology. TPSA grew out of the Theoretical Physics Student Association of Ireland, and the researchers behind the COTHROM project are current or former Trinity College Dublin students.

Notes to editors

- Research team: Ruaidhrí Campion, Joshua Cooney Mercadal, Casey Farren-Colloty, Matthew Fenlon, Eliza Somerville, Michael A. J. Mitchell.
- Research paper: *COTHROM: Algorithmic Redistricting of Irish Election Boundaries* (June 2026).
- The Cork figures are results from a proof-of-concept study to illustrate the method, and should not be taken as a proposed map.
- The Irish Times coverage: *TCD mathematicians develop a formula to solve constituency boundary conundrum*, Harry McGee (July 2026).

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